

Integrative analysis for sc-/sn-RNA-Seq Analysis in 10X Genomics data

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```
## The following objects are masked _by_ .GlobalEnv:
##
##   PCA_Feature_List_value, Regress_Cell_Cycle_value, assay,
##   genome_name, integration_method, nfeatures_value, root_dir
```

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DNB Bioinformatics Core Pipeline: Standard sc-/sn-RNA-Seq Analysis in 10X Genomics data

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1 Information about this notebook

Biological heterogeneity in single-cell RNA-seq data is often confounded by technical factors including sequencing depth. The number of molecules detected in each cell can vary significantly between cells, even within the same celltype. Interpretation of scRNA-seq data requires effective pre-processing and normalization to remove this technical variability (for more information, see also scRNA-seq Dataset Integration²).

Hence, integration of single-cell sequencing datasets, for example across experimental batches, donors, or conditions, is often an important step in scRNA-seq workflows. Integrative analysis can help to match shared cell types and states across datasets, which can boost statistical power, and most importantly, facilitate accurate comparative analysis across datasets. Here, we include the following integration methods for integrative analysis:

- (1) Seurat ‘anchor-based’³ integration workflow by Choudhary & Satija, 2022⁴,
- (2) Harmony⁵ integration workflow by Korsunsky et al., 2019⁶, and
- (3) Liger⁷ integration workflow by Welch et al., 2019⁸.

Different integration algorithms, such as Seurat, Harmony, and Liger, often use different numbers of dimensions. The optimal number of dimensions depends on the specific algorithm and its associated parameters. While these algorithms generally consider similar variables, the mathematical approaches they use to process and transform those variables differ significantly, which means the same number of dimensions (or “resolutions”) doesn’t work equally well across all algorithms.

Even though they all refer to “resolution,” the effect of resolution can vary from one algorithm to another, and the maximum number of dimensions they can effectively use may differ. Based on our experience, we recommend using 30 dimensions for Seurat, 30 for Harmony, and 20 for Liger. These settings tend to produce the most consistent integration results without unnecessarily increasing computational load.

2 Set up

```
attach(params)
```

```
## The following objects are masked _by_ .GlobalEnv:  
##  
##   PCA_Feature_List_value, Regress_Cell_Cycle_value, assay,  
##   genome_name, integration_method, nfeatures_value, root_dir
```

```
suppressPackageStartupMessages({  
  library(future)  
  library(tidyverse)  
  library(patchwork)  
  library(Seurat)  
  library(SeuratObject)  
  library(harmony)
```

²<https://www.singlecellcourse.org/scrna-seq-dataset-integration.html#seurat-v3-3-vs-5-10k-pbmc>

³https://satijalab.org/seurat/articles/seurat5_integration

⁴<https://doi.org/10.1186/s13059-021-02584-9>

⁵<https://portals.broadinstitute.org/harmony/>

⁶<https://www.nature.com/articles/s41592-019-0619-0>

⁷https://welch-lab.github.io/liger/articles/liger_with_seurat.html

⁸<https://doi.org/10.1016/j.cell.2019.05.006>

```

library(rliger)
library(RcppPlanc)
library(SeuratWrappers)
library(scooter)
library(reshape2)
library(RColorBrewer)
library(knitr)

# Evaluate Seurat R expressions asynchronously when possible using future package
options(future.globals.maxSize = future_globals_value)
plan(multisession, workers = parallelly::availableCores())}

```

3 Directories and paths to file Inputs/Outputs

```

#analysis_dir <- file.path(root_dir, "analyses", "integrative-analysis")
#data_dir <- file.path(root_dir, "analyses", "upstream-analysis", "results", "04_Filter_obj")
#upstream_analysis_dir <- file.path(root_dir, "analyses", "upstream-analysis")

# Input files
#data_file <- file.path(data_dir, "seurat_obj_merged_filtered.rds")

# Create results_dir
#results_dir <- file.path(analysis_dir, "results")
#if (!dir.exists(results_dir)) {
#  dir.create(results_dir)}

# Create plots directory
#report_dir <- file.path(analysis_dir, "plots")
#if (!dir.exists(report_dir)) {
#  dir.create(report_dir)}

#plots_dir <- file.path(report_dir, glue::glue("plots_{integration_method}"))
#if (!dir.exists(plots_dir)) {
#  dir.create(plots_dir)}

source(paste0(root_dir, "/figures/scripts/theme_plot.R"))
#source(paste0(analysis_dir, "/util/function-samples-integrate.R"))
#source(paste0(analysis_dir, "/util/custom-seurat-functions.R"))
#source(paste0(upstream_analysis_dir, "/util/function-process-Seurat.R"))

source(paste0(module_dir, "/util/function-samples-integrate.R"))
source(paste0(module_dir, "/util/custom-seurat-functions.R"))
source(paste0(module_dir, "/util/function-process-Seurat.R"))

```

4 Read seurat object

First, we will use the `seurat_obj_merged_filtered.rds` object as generated from the pipeline in the `upstream-analysis` module. This object contains all samples of the project merged.


```
merged_obj <- readRDS(data_file)
```

5 Seurat ‘anchor-based’ integration workflow

Seurat integration creates a unified object that contains both original data (‘RNA’ assay) as well as integrated data (‘integrated’ assay).

5.1 Normal-sized datasets

SCT method (or `sctransform`) is a modeling framework for the normalization and variance stabilization of molecular count data from scRNA-seq experiments. This procedure omits the need for heuristic steps including pseudocount addition or log-transformation and improves common downstream analytical tasks such as variable gene selection, dimensional reduction, and differential expression. More specifically, the `sctransform` approach utilizes the Pearson residuals from negative binomial regression as input to standard dimensional reduction techniques.

5.2 Large-sized datasets

For very large datasets, the standard integration workflow can sometimes be prohibitively computationally expensive. In addition, Seurat and a lot of integration algorithms are designed to integrate data up to a maximum amount. Here, we use the `big_data` parameter as an option for integration of extremely large datasets. For example, 1.3 millions cells are too much and so this option exist for those datasets. The problem comes down to infinite matrix sizes and memory issues, no computer to data can compute this many variables so new algorithms exist to get around that through various methods. For more information, see Seurat v4 large data integration⁹ and Seurat v5 large data integration¹⁰.

A `reference_list`¹¹ can be used when conducting Seurat integration which help with speeding up integration.

Moreover, we improve efficiency and runtimes by employing (1) Reciprocal PCA (RPCA) and (2) Reference-based integration. The main efficiency improvements are gained in `FindIntegrationAnchors()`. First, we use reciprocal PCA (RPCA), to identify an effective space in which to find anchors. When determining anchors between any two datasets using reciprocal PCA, we project each dataset into the others PCA space and constrain the anchors by the same mutual neighborhood requirement. All downstream integration steps remain the same and we are able to ‘correct’ (or harmonize) the datasets.

Additionally, we use reference-based integration. In the standard workflow, we identify anchors between all pairs of datasets. While this gives datasets equal weight in downstream integration, it can also become computationally intensive. For example when integrating 10 different datasets, we perform 45 different pairwise comparisons. As an alternative, we introduce here the possibility of specifying one or more of the datasets as the ‘reference’ for integrated analysis, with the remainder designated as ‘query’ datasets. In this workflow, we do not identify anchors between pairs of query datasets, reducing the number of comparisons. For example, when integrating 10 datasets with one specified as a reference, we perform only 9 comparisons. Reference-based integration can be applied to either log-normalized or SCTransform-normalized datasets.

Note that the anchor-based RPCA integration represents a faster and more conservative (less correction) method for integration.

⁹https://satijalab.org/seurat/archive/v4.3/integration_large_datasets

¹⁰https://satijalab.org/seurat/articles/seurat5_sketch_analysis

¹¹<https://satijalab.org/seurat/reference/findintegrationanchors#examples>

```

# We will use the Seurat 'anchor-based' workflow, if defined in `params`.
if (use_seurat_integration == "YES"){
  print_message <- "we use the Seurat 'anchor-based' workflow"

  # Make seurat_list
  seurat_obj_list <- list()
  seurat_obj_list <- SplitObject(merged_obj, split.by = "orig.ident")

  # Run function
  seurat_obj <- seurat_integration(seurat_obj_list = seurat_obj_list,
                                   nfeatures_value = nfeatures_value,
                                   num_dim_seurat = num_dim_seurat,
                                   num_dim_seurat_integration = num_dim_seurat_integration,
                                   big_data = big_data_value,
                                   Genome = genome_name,
                                   Regress_Cell_Cycle = Regress_Cell_Cycle_value,
                                   reference_list = reference_list_value,
                                   PCA_Feature_List = PCA_Feature_List_value)
} else {
  print_message <- "we will skip usage of the Seurat 'anchor-based' workflow"
}

```

Here, we will skip usage of the Seurat 'anchor-based' workflow.

6 Harmony integration workflow

Harmony is a general-purpose R package with an efficient algorithm for integrating multiple data sets. It is especially useful for large single-cell datasets such as single-cell RNA-seq. Harmony algorithm projects cells into a shared embedding in which cells group by cell type rather than dataset-specific conditions. Harmony simultaneously accounts for multiple experimental and biological factors. We will run Harmony within our Seurat workflow.

```

# We will use the Harmony integration workflow, if defined in `params`.
if (use_harmony_integration == "YES"){
  print_message <- "we use the Harmony integration workflow"
  seurat_obj <- harmony_integration(seurat_obj = merged_obj,
                                    variables_to_integrate = variable_value,
                                    num_dim = num_dim_harmony)

  # Plot
  name <- paste0(plots_dir, "/", glue::glue("plot_{integration_method}.png"))
  p1 <- DimPlot(object = seurat_obj, reduction = glue::glue("{integration_method}"), pt.size = 100)
  p2 <- VlnPlot(object = seurat_obj, features = "harmony_1", group.by = "orig.ident", pt.size = 100)
  print(patchwork::wrap_plots(list(p1, p2), nrow = 1))
  ggsave(file = name, width = 12, height = 5, device = "png")
} else {
  print_message <- "we will skip usage of the Harmony integration workflow"
}

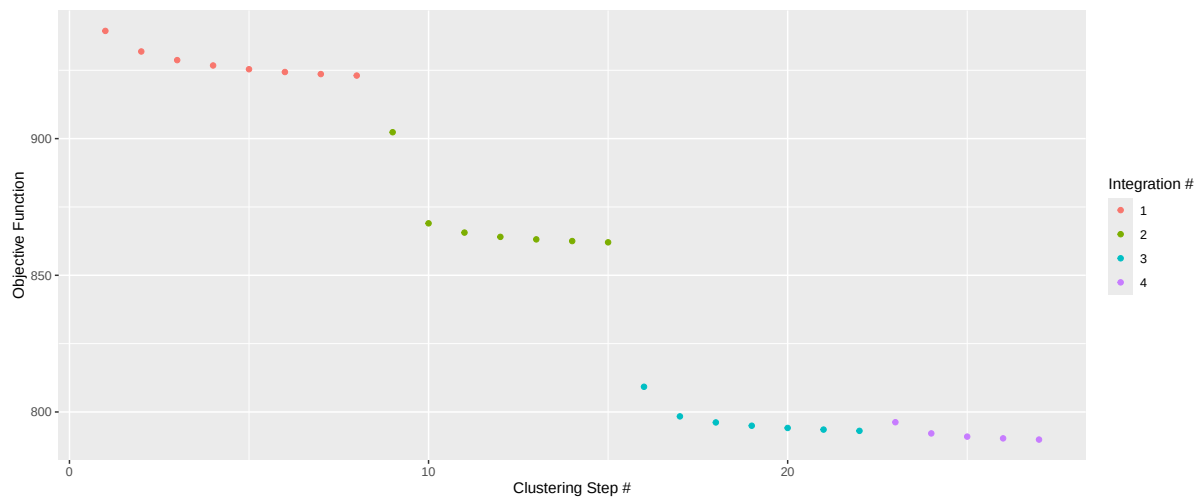
```

```
## Performing Harmony Integration
```

```
## Transposing data matrix
```

```
## Initializing state using k-means centroids initialization
```

```
## Harmony 1/10
## Harmony 2/10
## Harmony 3/10
## Harmony 4/10
## Harmony converged after 4 iterations
```



```
## Performing Uniform Manifold Approximation and Projection (UMAP) dimensional reduction te
## 19:30:07 UMAP embedding parameters a = 0.9922 b = 1.112
## 19:30:07 Read 51800 rows and found 30 numeric columns
## 19:30:07 Using Annoy for neighbor search, n_neighbors = 30
## 19:30:07 Building Annoy index with metric = cosine, n_trees = 50
## 0% 10 20 30 40 50 60 70 80 90 100%
## [----|----|----|----|----|----|----|----|----|----|
## *****|
## 19:30:14 Writing NN index file to temp file /lsf_tmp/264625707.tmpdir/RtmpbhtMvE/file2f9
## 19:30:14 Searching Annoy index using 16 threads, search_k = 3000
## 19:30:15 Annoy recall = 100%
## 19:30:16 Commencing smooth kNN distance calibration using 16 threads with target n_neigh
## 19:30:17 Initializing from normalized Laplacian + noise (using irlba)
## 19:30:18 Commencing optimization for 200 epochs, with 2200094 positive edges
## 19:30:39 Optimization finished
## Generate and add metadata
##
##
## ===== saving metadata =====
##
##
##
## ===== saving metadata =====
```



```

# Plot
name <- paste0(plots_dir, "/", glue::glue("plot_{integration_method}.png"))
DimPlot(seurat_obj, reduction = glue::glue("{integration_method}"), group.by = c("orig.ident"),
ggsave(file = name, width = 6, height = 5, device = "png")

} else {
  print_message <- "we will skip usage of the Liger workflow"

```

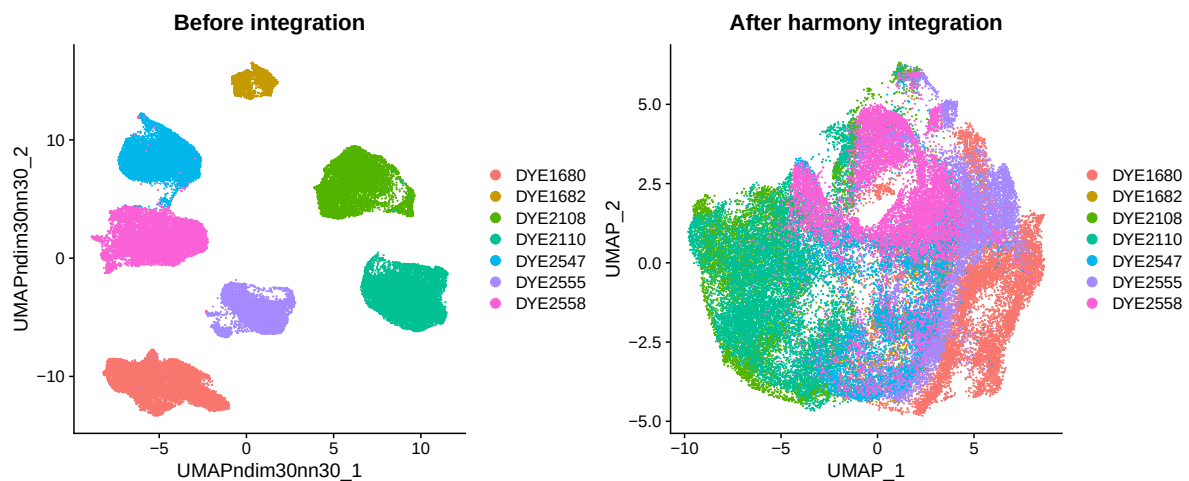
Here, we will skip usage of the Liger workflow.

8 Before vs After integration plot

```

# Plot
name <- paste0(plots_dir, "/", glue::glue("plot_unintegrated_vs_{integration_method}.png"))
p1 <- DimPlot(merged_obj, group.by = c("orig.ident")) + ggtitle("Before integration")
p2 <- DimPlot(seurat_obj, reduction = "umap", group.by = c("orig.ident")) + ggtitle(glue::glue("{integration_method}"))
patchwork::wrap_plots(list(p1, p2), nrow = 1)

```



```

ggsave(file = name, width = 12, height = 5, device = "png")

```

```

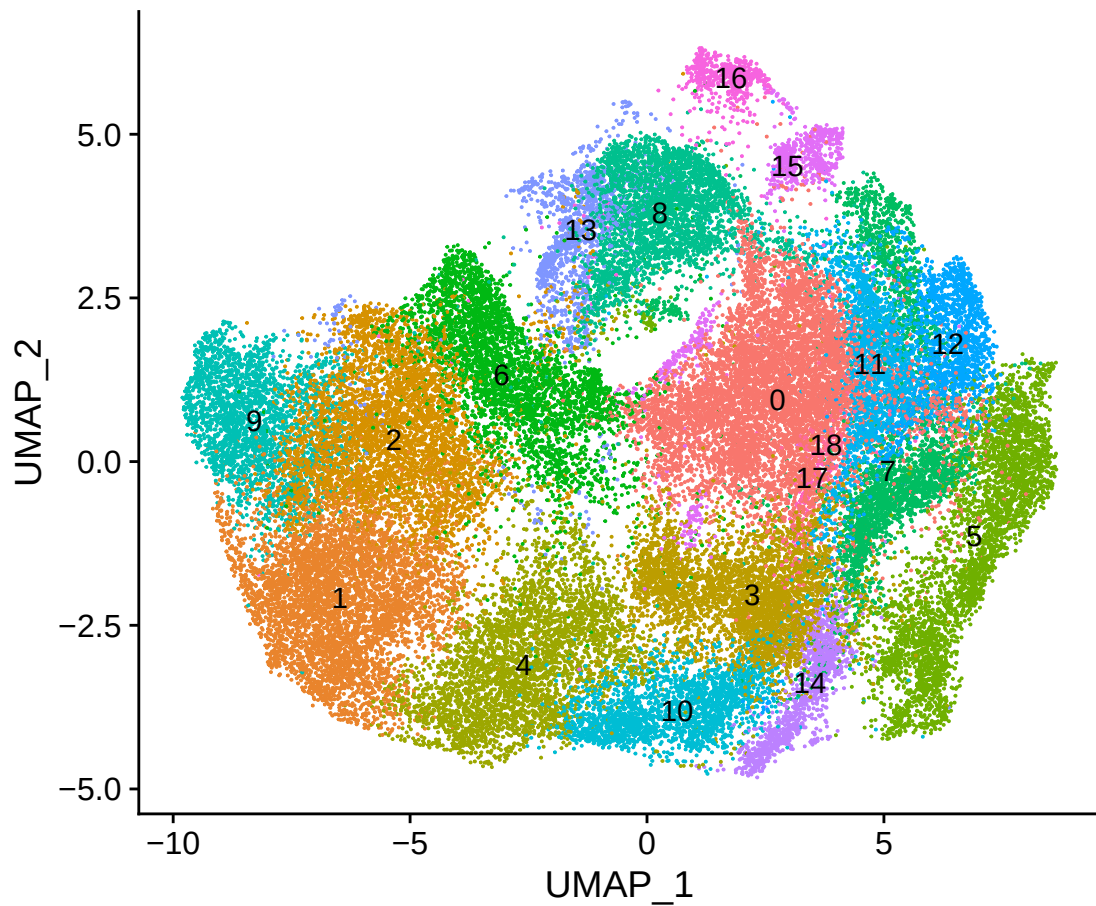
seurat_obj <- SetIdent(seurat_obj, value = "seurat_clusters")

```

```

# Plot
name <- paste0(plots_dir, "/", glue::glue("plot_clusters_{integration_method}.png"))
DimPlot(seurat_obj, reduction = "umap", label = TRUE) + NoLegend()

```

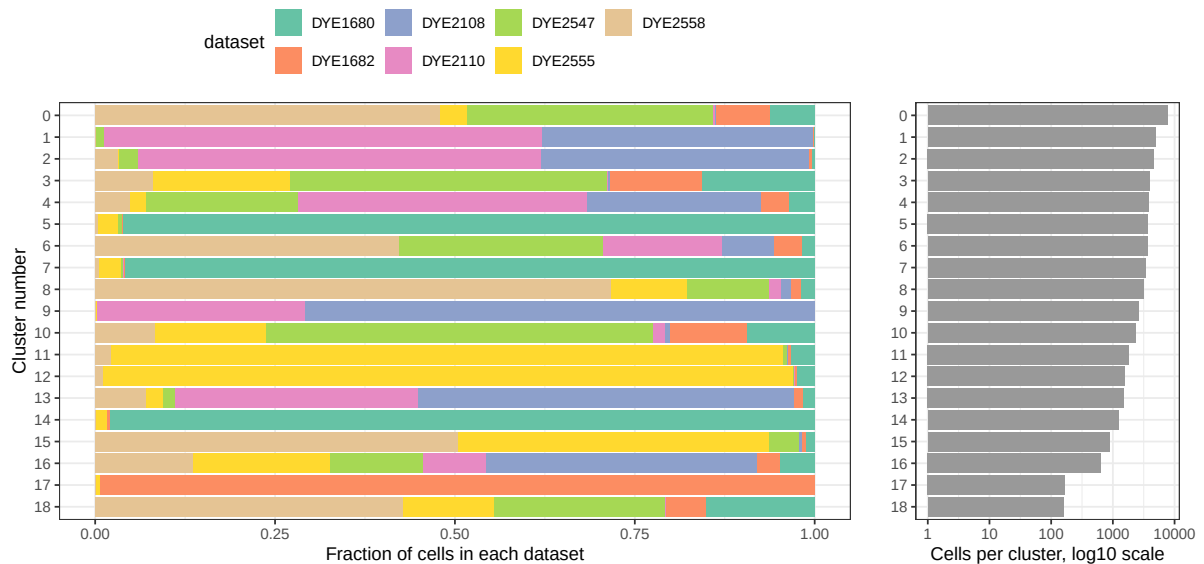


```
ggsave(file = name, width = 6, height = 5, device = "png")
```

```
# Plot
```

```
name <- paste0(plots_dir, "/", glue::glue("plot_fraction_of_cells_{integration_method}.png"))
plot_integrated_clusters(seurat_obj)
```

```
## Using cluster as id variables
```




```
ggsave(file = name, width = 10, height = 6, device = "png")
```

9 Save output files

```
# Identify columns with a '.1' suffix
cols_to_remove <- grep("\\.1$", colnames(seurat_obj@meta.data), value = TRUE)

# Exclude columns that match the specific patterns (e.g., {assay}_snn_res.0.1, {assay}_snn_res.1.0)
cols_to_remove <- cols_to_remove[!grepl(glue::glue("^{assay}_snn_res\\.0\\.1$"), cols_to_remove) &&
!grepl(glue::glue("^{assay}_snn_res\\.1$"), cols_to_remove) &&
!grepl(glue::glue("^{assay}_snn_res\\.10$"), cols_to_remove)]

# Remove the columns
seurat_obj@meta.data <- seurat_obj@meta.data[, !colnames(seurat_obj@meta.data) %in% cols_to_remove]
head(seurat_obj@meta.data)
```

```
##               orig.ident nCount_RNA nFeature_RNA percent.mito
## DYE1680:AAACCCAAGAGTCAGC   DYE1680      35918         5726      2.809
## DYE1680:AAACCCAAGATACCAA   DYE1680      25669         5653      1.963
## DYE1680:AAACCCAAGCTTCTAG   DYE1680      34294         6052      1.773
## DYE1680:AAACCCACACTGGATT   DYE1680      36570         6302      1.602
## DYE1680:AAACCCACAGAAGTGC   DYE1680      14405         3982      3.547
## DYE1680:AAACCCAGTAAGAACT   DYE1680      33962         7176      3.578
##
##               log10GenesPerUMI      ID      SAMPLE
## DYE1680:AAACCCAAGAGTCAGC      0.8249383 DYE1680 2664137_DYE1680
## DYE1680:AAACCCAAGATACCAA      0.8509710 DYE1680 2664137_DYE1680
## DYE1680:AAACCCAAGCTTCTAG      0.8338957 DYE1680 2664137_DYE1680
## DYE1680:AAACCCACACTGGATT      0.8326483 DYE1680 2664137_DYE1680
## DYE1680:AAACCCACAGAAGTGC      0.8657184 DYE1680 2664137_DYE1680
## DYE1680:AAACCCAGTAAGAACT      0.8510016 DYE1680 2664137_DYE1680
##
## DYE1680:AAACCCAAGAGTCAGC /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/
1/2-2261003/
## DYE1680:AAACCCAAGATACCAA /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/
1/2-2261003/
## DYE1680:AAACCCAAGCTTCTAG /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/
1/2-2261003/
## DYE1680:AAACCCACACTGGATT /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/
1/2-2261003/
## DYE1680:AAACCCACAGAAGTGC /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/
1/2-2261003/
## DYE1680:AAACCCAGTAAGAACT /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/
1/2-2261003/
##
##               SJUID      SJID      cancer_type
## DYE1680:AAACCCAAGAGTCAGC SJHBBBD73HK SJRHB000026 Rhabdomyosarcoma
## DYE1680:AAACCCAAGATACCAA SJHBBBD73HK SJRHB000026 Rhabdomyosarcoma
## DYE1680:AAACCCAAGCTTCTAG SJHBBBD73HK SJRHB000026 Rhabdomyosarcoma
## DYE1680:AAACCCACACTGGATT SJHBBBD73HK SJRHB000026 Rhabdomyosarcoma
## DYE1680:AAACCCACAGAAGTGC SJHBBBD73HK SJRHB000026 Rhabdomyosarcoma
## DYE1680:AAACCCAGTAAGAACT SJHBBBD73HK SJRHB000026 Rhabdomyosarcoma
```

##	Subject_Disease_Code	Subject_Disease_Subtype_Code		
## DYE1680:AAACCCAAGAGTCAGC	RMS	ERMS		
## DYE1680:AAACCCAAGATACCAA	RMS	ERMS		
## DYE1680:AAACCCAAGCTTCTAG	RMS	ERMS		
## DYE1680:AAACCCACACTGGATT	RMS	ERMS		
## DYE1680:AAACCCACAGAAGTGC	RMS	ERMS		
## DYE1680:AAACCCAGTAAGAACT	RMS	ERMS		
##	tissue_type	assay_technology_updated	assay_technology	
## DYE1680:AAACCCAAGAGTCAGC	CCLF_cells	10x_scrNAseq	snRNA seq	
## DYE1680:AAACCCAAGATACCAA	CCLF_cells	10x_scrNAseq	snRNA seq	
## DYE1680:AAACCCAAGCTTCTAG	CCLF_cells	10x_scrNAseq	snRNA seq	
## DYE1680:AAACCCACACTGGATT	CCLF_cells	10x_scrNAseq	snRNA seq	
## DYE1680:AAACCCACAGAAGTGC	CCLF_cells	10x_scrNAseq	snRNA seq	
## DYE1680:AAACCCAGTAAGAACT	CCLF_cells	10x_scrNAseq	snRNA seq	
##	historic_data	RPMI	genome_reference_updated_code	
## DYE1680:AAACCCAAGAGTCAGC	no	no	human-genome	
## DYE1680:AAACCCAAGATACCAA	no	no	human-genome	
## DYE1680:AAACCCAAGCTTCTAG	no	no	human-genome	
## DYE1680:AAACCCACACTGGATT	no	no	human-genome	
## DYE1680:AAACCCACAGAAGTGC	no	no	human-genome	
## DYE1680:AAACCCAGTAAGAACT	no	no	human-genome	
##	genome_reference_updated	genome_reference		
## DYE1680:AAACCCAAGAGTCAGC	dual-genome	dual-genome		
## DYE1680:AAACCCAAGATACCAA	dual-genome	dual-genome		
## DYE1680:AAACCCAAGCTTCTAG	dual-genome	dual-genome		
## DYE1680:AAACCCACACTGGATT	dual-genome	dual-genome		
## DYE1680:AAACCCACAGAAGTGC	dual-genome	dual-genome		
## DYE1680:AAACCCAGTAAGAACT	dual-genome	dual-genome		
##				
## DYE1680:AAACCCAAGAGTCAGC	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/			
## DYE1680:AAACCCAAGATACCAA	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/			
## DYE1680:AAACCCAAGCTTCTAG	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/			
## DYE1680:AAACCCACACTGGATT	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/			
## DYE1680:AAACCCACAGAAGTGC	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/			
## DYE1680:AAACCCAGTAAGAACT	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/			
##	DYExxxx	DYE_Melody_unsure	DYExxxx_corrected	
## DYE1680:AAACCCAAGAGTCAGC	DYE1680	<NA>	DYE1680	
## DYE1680:AAACCCAAGATACCAA	DYE1680	<NA>	DYE1680	
## DYE1680:AAACCCAAGCTTCTAG	DYE1680	<NA>	DYE1680	
## DYE1680:AAACCCACACTGGATT	DYE1680	<NA>	DYE1680	
## DYE1680:AAACCCACAGAAGTGC	DYE1680	<NA>	DYE1680	
## DYE1680:AAACCCAGTAAGAACT	DYE1680	<NA>	DYE1680	
##	SRM_updated	SRM	SRM_Sample_ID_updated	SRM_Sample_ID
## DYE1680:AAACCCAAGAGTCAGC	311618	311618	2664137	2664137
## DYE1680:AAACCCAAGATACCAA	311618	311618	2664137	2664137

## DYE1680:AAACCCAAGCTTCTAG	311618	311618	2664137	2664137
## DYE1680:AAACCCACACTGGATT	311618	311618	2664137	2664137
## DYE1680:AAACCCACAGAAGTGC	311618	311618	2664137	2664137
## DYE1680:AAACCCAGTAAGAACT	311618	311618	2664137	2664137
##	Experimental_Sample sample_3D			
## DYE1680:AAACCCAAGAGTCAGC	SJ1001	No		
## DYE1680:AAACCCAAGATACCAA	SJ1001	No		
## DYE1680:AAACCCAAGCTTCTAG	SJ1001	No		
## DYE1680:AAACCCACACTGGATT	SJ1001	No		
## DYE1680:AAACCCACAGAAGTGC	SJ1001	No		
## DYE1680:AAACCCAGTAAGAACT	SJ1001	No		
##	sample_2D.NOT.NEEDED.2025.04.10 Tissue_Bank_ID			
## DYE1680:AAACCCAAGAGTCAGC		No	TB-12-1442	
## DYE1680:AAACCCAAGATACCAA		No	TB-12-1442	
## DYE1680:AAACCCAAGCTTCTAG		No	TB-12-1442	
## DYE1680:AAACCCACACTGGATT		No	TB-12-1442	
## DYE1680:AAACCCACAGAAGTGC		No	TB-12-1442	
## DYE1680:AAACCCAGTAAGAACT		No	TB-12-1442	
##	Xenograft_CompBio_ID CSTN_MAST_ID Date			
## DYE1680:AAACCCAAGAGTCAGC	SJRHB000026_X1	MAST39	2023-10-23T00:00:00Z	
## DYE1680:AAACCCAAGATACCAA	SJRHB000026_X1	MAST39	2023-10-23T00:00:00Z	
## DYE1680:AAACCCAAGCTTCTAG	SJRHB000026_X1	MAST39	2023-10-23T00:00:00Z	
## DYE1680:AAACCCACACTGGATT	SJRHB000026_X1	MAST39	2023-10-23T00:00:00Z	
## DYE1680:AAACCCACAGAAGTGC	SJRHB000026_X1	MAST39	2023-10-23T00:00:00Z	
## DYE1680:AAACCCAGTAAGAACT	SJRHB000026_X1	MAST39	2023-10-23T00:00:00Z	
##	SRM_Submission_Information			
## DYE1680:AAACCCAAGAGTCAGC	<NA>			
## DYE1680:AAACCCAAGATACCAA	<NA>			
## DYE1680:AAACCCAAGCTTCTAG	<NA>			
## DYE1680:AAACCCACACTGGATT	<NA>			
## DYE1680:AAACCCACAGAAGTGC	<NA>			
## DYE1680:AAACCCAGTAAGAACT	<NA>			
##	Compbioname_for_publications Comp_Bio_Sample_Name			
## DYE1680:AAACCCAAGAGTCAGC	SJRHB000026_C5	SJRHB000026_C5-		
SJ1001T_2				
## DYE1680:AAACCCAAGATACCAA	SJRHB000026_C5	SJRHB000026_C5-		
SJ1001T_2				
## DYE1680:AAACCCAAGCTTCTAG	SJRHB000026_C5	SJRHB000026_C5-		
SJ1001T_2				
## DYE1680:AAACCCACACTGGATT	SJRHB000026_C5	SJRHB000026_C5-		
SJ1001T_2				
## DYE1680:AAACCCACAGAAGTGC	SJRHB000026_C5	SJRHB000026_C5-		
SJ1001T_2				
## DYE1680:AAACCCAGTAAGAACT	SJRHB000026_C5	SJRHB000026_C5-		
SJ1001T_2				
##	Comp_Bio_Subject_ID Preferred_Sample_ID			
## DYE1680:AAACCCAAGAGTCAGC	SJ000026	SJ1001T_2		
## DYE1680:AAACCCAAGATACCAA	SJ000026	SJ1001T_2		
## DYE1680:AAACCCAAGCTTCTAG	SJ000026	SJ1001T_2		
## DYE1680:AAACCCACACTGGATT	SJ000026	SJ1001T_2		
## DYE1680:AAACCCACAGAAGTGC	SJ000026	SJ1001T_2		

##	DYE1680:AAACCCAGTAAGAACT	SJ000026	SJ1001T_2		
##		Sample_Type_Code	Prior_Comp_Bio_ID	PATH	NOTES
##	DYE1680:AAACCCAAGAGTCAGC	C	SJRHB000026	<NA>	NA
##	DYE1680:AAACCCAAGATACCAA	C	SJRHB000026	<NA>	NA
##	DYE1680:AAACCCAAGCTTCTAG	C	SJRHB000026	<NA>	NA
##	DYE1680:AAACCCACACTGGATT	C	SJRHB000026	<NA>	NA
##	DYE1680:AAACCCACAGAAGTGC	C	SJRHB000026	<NA>	NA
##	DYE1680:AAACCCAGTAAGAACT	C	SJRHB000026	<NA>	NA
##		match_SJID_RPMI	ID_with_sample_swap		
##	DYE1680:AAACCCAAGAGTCAGC	SJRHB000026_no	DYE1680		
##	DYE1680:AAACCCAAGATACCAA	SJRHB000026_no	DYE1680		
##	DYE1680:AAACCCAAGCTTCTAG	SJRHB000026_no	DYE1680		
##	DYE1680:AAACCCACACTGGATT	SJRHB000026_no	DYE1680		
##	DYE1680:AAACCCACAGAAGTGC	SJRHB000026_no	DYE1680		
##	DYE1680:AAACCCAGTAAGAACT	SJRHB000026_no	DYE1680		
##		SAMPLE_with_sample_swap			
##	DYE1680:AAACCCAAGAGTCAGC	2664137_DYE1680			
##	DYE1680:AAACCCAAGATACCAA	2664137_DYE1680			
##	DYE1680:AAACCCAAGCTTCTAG	2664137_DYE1680			
##	DYE1680:AAACCCACACTGGATT	2664137_DYE1680			
##	DYE1680:AAACCCACAGAAGTGC	2664137_DYE1680			
##	DYE1680:AAACCCAGTAAGAACT	2664137_DYE1680			
##					
##	DYE1680:AAACCCAAGAGTCAGC	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/			
##	DYE1680:AAACCCAAGATACCAA	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/			
##	DYE1680:AAACCCAAGCTTCTAG	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/			
##	DYE1680:AAACCCACACTGGATT	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/			
##	DYE1680:AAACCCACAGAAGTGC	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/			
##	DYE1680:AAACCCAGTAAGAACT	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/			
##		S.Score	G2M.Score	Phase	UMAP_1 UMAP_2
##	DYE1680:AAACCCAAGAGTCAGC	-0.1902376	-0.3227688	G1	-1.4770168 9.253440
##	DYE1680:AAACCCAAGATACCAA	-0.1182429	-0.2969047	G1	-3.4309131 9.242017
##	DYE1680:AAACCCAAGCTTCTAG	-0.2180694	-0.3405832	G1	-1.3636737 5.705616
##	DYE1680:AAACCCACACTGGATT	-0.2376908	-0.3945082	G1	-2.9635712 9.708883
##	DYE1680:AAACCCACAGAAGTGC	-0.1505212	-0.1332644	G1	-1.0462393 7.168882
##	DYE1680:AAACCCAGTAAGAACT	0.4046701	-0.1800174	S	0.4893034 7.784715
##		PC_1	PC_2	PC_3	PC_4 PC_5
##	DYE1680:AAACCCAAGAGTCAGC	-16.39490	-0.06040939	8.992026	-5.282849 2.5461725
##	DYE1680:AAACCCAAGATACCAA	-16.64311	-0.38841290	8.531777	-4.574383 -0.1577565
##	DYE1680:AAACCCAAGCTTCTAG	-17.02470	-0.26009097	9.657486	-4.846276 2.6670944
##	DYE1680:AAACCCACACTGGATT	-15.89784	0.65302425	9.408316	-4.259913 -0.3211939
##	DYE1680:AAACCCACAGAAGTGC	-15.76642	0.20835809	7.609702	-3.459250 2.9684272
##	DYE1680:AAACCCAGTAAGAACT	-11.64230	1.66521080	7.193732	-2.240468 2.9785559
##		PC_6	PC_7	PC_8	PC_9
##	DYE1680:AAACCCAAGAGTCAGC	6.56743106	5.4881562	4.7613897	-1.7865470

## DYE1680:AAACCCAAGATACCAA	3.63397381	-0.4869838	-0.6133165	0.4781322	
## DYE1680:AAACCCAAGCTTCTAG	8.97802837	4.5605539	2.0849652	-1.6105814	
## DYE1680:AAACCCACACTGGATT	-0.05474395	-2.2903917	3.7300108	-0.5776052	
## DYE1680:AAACCCACAGAAGTGC	9.52270257	4.2045560	0.8241562	-1.4506361	
## DYE1680:AAACCCAGTAAGAACT	7.74266906	4.9746429	-0.1754950	0.8894317	
##	PC_10	PC_11	PC_12	PC_13	PC_14
## DYE1680:AAACCCAAGAGTCAGC	0.4089495	0.5652035	-0.3806822	-1.011628	1.4937597
## DYE1680:AAACCCAAGATACCAA	1.9618030	1.2614855	1.0912301	-1.861864	2.1052791
## DYE1680:AAACCCAAGCTTCTAG	2.0010723	0.1003519	0.4319290	-2.877628	3.0628591
## DYE1680:AAACCCACACTGGATT	1.5921441	-1.2365924	-0.7285866	-2.319632	2.5677598
## DYE1680:AAACCCACAGAAGTGC	0.7336166	0.9973878	0.6110972	-1.937815	0.2880151
## DYE1680:AAACCCAGTAAGAACT	-1.4764380	-1.9228858	-3.1487889	2.355300	0.5375213
##	PC_15	PC_16	PC_17	PC_18	
## DYE1680:AAACCCAAGAGTCAGC	-0.6740777	-1.26045048	-1.1002345	5.6047590	
## DYE1680:AAACCCAAGATACCAA	2.1245733	0.53280848	-2.1309847	6.0946147	
## DYE1680:AAACCCAAGCTTCTAG	0.6736039	0.05249832	-1.3541124	4.7995531	
## DYE1680:AAACCCACACTGGATT	-1.4860827	-1.62879851	-2.3296588	4.5801502	
## DYE1680:AAACCCACAGAAGTGC	-0.3293910	1.34307035	-0.6013391	-0.8125743	
## DYE1680:AAACCCAGTAAGAACT	0.9685769	-2.24292698	0.8368750	-3.2915569	
##	PC_19	PC_20	PC_21	PC_22	PC_23
## DYE1680:AAACCCAAGAGTCAGC	2.8967472	2.553915	-1.1964531	-1.7446081	-
	0.38584092				
## DYE1680:AAACCCAAGATACCAA	-0.1098369	1.302813	0.5558706	0.5823816	0.36094618
## DYE1680:AAACCCAAGCTTCTAG	1.0072936	1.547666	-0.3104649	-0.8433231	0.18960913
## DYE1680:AAACCCACACTGGATT	1.3088686	1.924711	-1.7640707	-2.5096479	0.08471221
## DYE1680:AAACCCACAGAAGTGC	-0.5978964	-2.268368	0.8100166	0.8282191	0.05591339
## DYE1680:AAACCCAGTAAGAACT	-0.9768113	0.543290	0.8136991	2.9906515	-
	0.51044236				
##	PC_24	PC_25	PC_26	PC_27	PC_28
## DYE1680:AAACCCAAGAGTCAGC	0.26184242	0.2345830	2.535177	0.55438094	5.646553
## DYE1680:AAACCCAAGATACCAA	0.07283511	-2.0171602	-2.480327	-0.14655514	-
	5.756900				
## DYE1680:AAACCCAAGCTTCTAG	0.51470577	-1.2350765	1.248190	1.13687684	2.915157
## DYE1680:AAACCCACACTGGATT	0.20947787	-1.5374189	2.780055	0.00924215	2.388767
## DYE1680:AAACCCACAGAAGTGC	0.60514027	-0.2262589	-1.149257	2.78306784	-
	2.389567				
## DYE1680:AAACCCAGTAAGAACT	-0.59741445	5.0010707	-0.746392	-1.64070414	-
	5.421385				
##	PC_29	PC_30	RNA_snn_res.0	seurat_clusters	
## DYE1680:AAACCCAAGAGTCAGC	3.754707	-1.2801447	1		7
## DYE1680:AAACCCAAGATACCAA	-4.257351	1.1886556	1		7
## DYE1680:AAACCCAAGCTTCTAG	1.994136	-0.1955019	1		7
## DYE1680:AAACCCACACTGGATT	1.642289	0.0246024	1		7
## DYE1680:AAACCCACAGAAGTGC	-3.795585	3.8918452	1		7
## DYE1680:AAACCCAGTAAGAACT	-6.199011	1.9952317	1		5
##	RNA_snn_res.0.1	RNA_snn_res.0.2	RNA_snn_res.0.3		
## DYE1680:AAACCCAAGAGTCAGC	1	1			8
## DYE1680:AAACCCAAGATACCAA	1	1			8
## DYE1680:AAACCCAAGCTTCTAG	1	1			1
## DYE1680:AAACCCACACTGGATT	1	1			8
## DYE1680:AAACCCACAGAAGTGC	1	1			1

## DYE1680:AAACCCAGTAAGAACT	1	1	1
##	RNA_snn_res.0.4	RNA_snn_res.0.5	RNA_snn_res.0.6
## DYE1680:AAACCCAAGAGTCAGC	8	9	7
## DYE1680:AAACCCAAGATACCAA	8	9	7
## DYE1680:AAACCCAAGCTTCTAG	1	9	7
## DYE1680:AAACCCACACTGGATT	8	9	7
## DYE1680:AAACCCACAGAAGTGC	8	9	7
## DYE1680:AAACCCAGTAAGAACT	10	8	10
##	RNA_snn_res.0.7	RNA_snn_res.0.8	RNA_snn_res.0.9
## DYE1680:AAACCCAAGAGTCAGC	8	7	9
## DYE1680:AAACCCAAGATACCAA	8	7	9
## DYE1680:AAACCCAAGCTTCTAG	8	7	9
## DYE1680:AAACCCACACTGGATT	8	7	9
## DYE1680:AAACCCACAGAAGTGC	8	7	9
## DYE1680:AAACCCAGTAAGAACT	11	5	13
##	RNA_snn_res.1	RNA_snn_res.2	RNA_snn_res.3
## DYE1680:AAACCCAAGAGTCAGC	10	27	26
## DYE1680:AAACCCAAGATACCAA	10	13	20
## DYE1680:AAACCCAAGCTTCTAG	10	27	26
## DYE1680:AAACCCACACTGGATT	10	27	26
## DYE1680:AAACCCACAGAAGTGC	10	13	39
## DYE1680:AAACCCAGTAAGAACT	12	11	39
##	RNA_snn_res.4	RNA_snn_res.5	RNA_snn_res.6
## DYE1680:AAACCCAAGAGTCAGC	28	17	15
## DYE1680:AAACCCAAGATACCAA	16	9	7
## DYE1680:AAACCCAAGCTTCTAG	28	17	15
## DYE1680:AAACCCACACTGGATT	28	17	15
## DYE1680:AAACCCACAGAAGTGC	44	52	52
## DYE1680:AAACCCAGTAAGAACT	44	62	82
##	RNA_snn_res.7	RNA_snn_res.8	RNA_snn_res.9
## DYE1680:AAACCCAAGAGTCAGC	11	4	6
## DYE1680:AAACCCAAGATACCAA	3	80	57
## DYE1680:AAACCCAAGCTTCTAG	11	4	6
## DYE1680:AAACCCACACTGGATT	11	4	6
## DYE1680:AAACCCACAGAAGTGC	47	51	32
## DYE1680:AAACCCAGTAAGAACT	95	104	95
##	RNA_snn_res.10	orig.ident.1	nCount_RNA.1
## DYE1680:AAACCCAAGAGTCAGC	2	DYE1680	35918
## DYE1680:AAACCCAAGATACCAA	58	DYE1680	25669
## DYE1680:AAACCCAAGCTTCTAG	2	DYE1680	34294
## DYE1680:AAACCCACACTGGATT	2	DYE1680	36570
## DYE1680:AAACCCACAGAAGTGC	29	DYE1680	14405
## DYE1680:AAACCCAGTAAGAACT	124	DYE1680	33962
##	nFeature_RNA.1	percent.mito.1	log10GenesPerUMI.1
## DYE1680:AAACCCAAGAGTCAGC	5726	2.809	0.8249383
## DYE1680:AAACCCAAGATACCAA	5653	1.963	0.8509710
## DYE1680:AAACCCAAGCTTCTAG	6052	1.773	0.8338957
## DYE1680:AAACCCACACTGGATT	6302	1.602	0.8326483
## DYE1680:AAACCCACAGAAGTGC	3982	3.547	0.8657184
## DYE1680:AAACCCAGTAAGAACT	7176	3.578	0.8510016
##	ID.1	SAMPLE.1	

```

## DYE1680:AAACCCAAGAGTCAGC DYE1680 2664137_DYE1680
## DYE1680:AAACCCAAGATACCAA DYE1680 2664137_DYE1680
## DYE1680:AAACCCAAGCTTCTAG DYE1680 2664137_DYE1680
## DYE1680:AAACCCACACTGGATT DYE1680 2664137_DYE1680
## DYE1680:AAACCCACAGAAGTGC DYE1680 2664137_DYE1680
## DYE1680:AAACCCAGTAAGAACT DYE1680 2664137_DYE1680
##
## DYE1680:AAACCCAAGAGTCAGC /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/1/2-2261003/
## DYE1680:AAACCCAAGATACCAA /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/1/2-2261003/
## DYE1680:AAACCCAAGCTTCTAG /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/1/2-2261003/
## DYE1680:AAACCCACACTGGATT /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/1/2-2261003/
## DYE1680:AAACCCACAGAAGTGC /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/1/2-2261003/
## DYE1680:AAACCCAGTAAGAACT /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/1/2-2261003/
##
## SJUID.1 SJID.1 cancer_type.1
## DYE1680:AAACCCAAGAGTCAGC SJHBBBD73HK SJRHB000026 Rhabdomyosarcoma
## DYE1680:AAACCCAAGATACCAA SJHBBBD73HK SJRHB000026 Rhabdomyosarcoma
## DYE1680:AAACCCAAGCTTCTAG SJHBBBD73HK SJRHB000026 Rhabdomyosarcoma
## DYE1680:AAACCCACACTGGATT SJHBBBD73HK SJRHB000026 Rhabdomyosarcoma
## DYE1680:AAACCCACAGAAGTGC SJHBBBD73HK SJRHB000026 Rhabdomyosarcoma
## DYE1680:AAACCCAGTAAGAACT SJHBBBD73HK SJRHB000026 Rhabdomyosarcoma
## Subject_Disease_Code.1 Subject_Disease_Subtype_Code.1
## DYE1680:AAACCCAAGAGTCAGC RMS ERMS
## DYE1680:AAACCCAAGATACCAA RMS ERMS
## DYE1680:AAACCCAAGCTTCTAG RMS ERMS
## DYE1680:AAACCCACACTGGATT RMS ERMS
## DYE1680:AAACCCACAGAAGTGC RMS ERMS
## DYE1680:AAACCCAGTAAGAACT RMS ERMS
## tissue_type.1 assay_technology_updated.1
## DYE1680:AAACCCAAGAGTCAGC CCLF_cells 10x_scRNAseq
## DYE1680:AAACCCAAGATACCAA CCLF_cells 10x_scRNAseq
## DYE1680:AAACCCAAGCTTCTAG CCLF_cells 10x_scRNAseq
## DYE1680:AAACCCACACTGGATT CCLF_cells 10x_scRNAseq
## DYE1680:AAACCCACAGAAGTGC CCLF_cells 10x_scRNAseq
## DYE1680:AAACCCAGTAAGAACT CCLF_cells 10x_scRNAseq
## assay_technology.1 historic_data.1 RPMI.1
## DYE1680:AAACCCAAGAGTCAGC snRNA seq no no
## DYE1680:AAACCCAAGATACCAA snRNA seq no no
## DYE1680:AAACCCAAGCTTCTAG snRNA seq no no
## DYE1680:AAACCCACACTGGATT snRNA seq no no
## DYE1680:AAACCCACAGAAGTGC snRNA seq no no
## DYE1680:AAACCCAGTAAGAACT snRNA seq no no
## genome_reference_updated_code.1
## DYE1680:AAACCCAAGAGTCAGC human-genome
## DYE1680:AAACCCAAGATACCAA human-genome
## DYE1680:AAACCCAAGCTTCTAG human-genome

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## DYE1680:AAACCCACACTGGATT	human-genome	
## DYE1680:AAACCCACAGAAGTGC	human-genome	
## DYE1680:AAACCCAGTAAGAACT	human-genome	
##	genome_reference_updated.1	genome_reference.1
## DYE1680:AAACCCAAGAGTCAGC	dual-genome	dual-genome
## DYE1680:AAACCCAAGATACCAA	dual-genome	dual-genome
## DYE1680:AAACCCAAGCTTCTAG	dual-genome	dual-genome
## DYE1680:AAACCCACACTGGATT	dual-genome	dual-genome
## DYE1680:AAACCCACAGAAGTGC	dual-genome	dual-genome
## DYE1680:AAACCCAGTAAGAACT	dual-genome	dual-genome
##		
## DYE1680:AAACCCAAGAGTCAGC	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina	
1/		
## DYE1680:AAACCCAAGATACCAA	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina	
1/		
## DYE1680:AAACCCAAGCTTCTAG	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina	
1/		
## DYE1680:AAACCCACACTGGATT	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina	
1/		
## DYE1680:AAACCCACAGAAGTGC	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina	
1/		
## DYE1680:AAACCCAGTAAGAACT	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina	
1/		
##	DYExxxx.1	DYE_Melody_unsure.1 DYExxxx_corrected.1
## DYE1680:AAACCCAAGAGTCAGC	DYE1680	<NA> DYE1680
## DYE1680:AAACCCAAGATACCAA	DYE1680	<NA> DYE1680
## DYE1680:AAACCCAAGCTTCTAG	DYE1680	<NA> DYE1680
## DYE1680:AAACCCACACTGGATT	DYE1680	<NA> DYE1680
## DYE1680:AAACCCACAGAAGTGC	DYE1680	<NA> DYE1680
## DYE1680:AAACCCAGTAAGAACT	DYE1680	<NA> DYE1680
##	SRM_updated.1	SRM.1 SRM_Sample_ID_updated.1
## DYE1680:AAACCCAAGAGTCAGC	311618	311618 2664137
## DYE1680:AAACCCAAGATACCAA	311618	311618 2664137
## DYE1680:AAACCCAAGCTTCTAG	311618	311618 2664137
## DYE1680:AAACCCACACTGGATT	311618	311618 2664137
## DYE1680:AAACCCACAGAAGTGC	311618	311618 2664137
## DYE1680:AAACCCAGTAAGAACT	311618	311618 2664137
##	SRM_Sample_ID.1	Experimental_Sample.1 sample_3D.1
## DYE1680:AAACCCAAGAGTCAGC	2664137	SJ1001 No
## DYE1680:AAACCCAAGATACCAA	2664137	SJ1001 No
## DYE1680:AAACCCAAGCTTCTAG	2664137	SJ1001 No
## DYE1680:AAACCCACACTGGATT	2664137	SJ1001 No
## DYE1680:AAACCCACAGAAGTGC	2664137	SJ1001 No
## DYE1680:AAACCCAGTAAGAACT	2664137	SJ1001 No
##	sample_2D.NOT.NEEDED.2025.04.10.1	Tissue_Bank_ID.1
## DYE1680:AAACCCAAGAGTCAGC	No	TB-12-1442
## DYE1680:AAACCCAAGATACCAA	No	TB-12-1442
## DYE1680:AAACCCAAGCTTCTAG	No	TB-12-1442
## DYE1680:AAACCCACACTGGATT	No	TB-12-1442
## DYE1680:AAACCCACAGAAGTGC	No	TB-12-1442
## DYE1680:AAACCCAGTAAGAACT	No	TB-12-1442

```

##                               Xenograft_CompBio_ID.1 CSTN_MAST_ID.1
## DYE1680:AAACCCAAGAGTCAGC      SJRHB000026_X1      MAST39
## DYE1680:AAACCCAAGATACCAA      SJRHB000026_X1      MAST39
## DYE1680:AAACCCAAGCTTCTAG      SJRHB000026_X1      MAST39
## DYE1680:AAACCCACACTGGATT      SJRHB000026_X1      MAST39
## DYE1680:AAACCCACAGAAGTGC      SJRHB000026_X1      MAST39
## DYE1680:AAACCCAGTAAGAACT      SJRHB000026_X1      MAST39
##                               Date.1 SRM_Submission_Information.1
## DYE1680:AAACCCAAGAGTCAGC 2023-10-23T00:00:00Z      <NA>
## DYE1680:AAACCCAAGATACCAA 2023-10-23T00:00:00Z      <NA>
## DYE1680:AAACCCAAGCTTCTAG 2023-10-23T00:00:00Z      <NA>
## DYE1680:AAACCCACACTGGATT 2023-10-23T00:00:00Z      <NA>
## DYE1680:AAACCCACAGAAGTGC 2023-10-23T00:00:00Z      <NA>
## DYE1680:AAACCCAGTAAGAACT 2023-10-23T00:00:00Z      <NA>
##                               Compbioname_for_publications.1
## DYE1680:AAACCCAAGAGTCAGC      SJRHB000026_C5
## DYE1680:AAACCCAAGATACCAA      SJRHB000026_C5
## DYE1680:AAACCCAAGCTTCTAG      SJRHB000026_C5
## DYE1680:AAACCCACACTGGATT      SJRHB000026_C5
## DYE1680:AAACCCACAGAAGTGC      SJRHB000026_C5
## DYE1680:AAACCCAGTAAGAACT      SJRHB000026_C5
##                               Comp_Bio_Sample_Name.1 Comp_Bio_Subject_ID.1
## DYE1680:AAACCCAAGAGTCAGC SJRHB000026_C5-SJ1001T_2      SJ000026
## DYE1680:AAACCCAAGATACCAA SJRHB000026_C5-SJ1001T_2      SJ000026
## DYE1680:AAACCCAAGCTTCTAG SJRHB000026_C5-SJ1001T_2      SJ000026
## DYE1680:AAACCCACACTGGATT SJRHB000026_C5-SJ1001T_2      SJ000026
## DYE1680:AAACCCACAGAAGTGC SJRHB000026_C5-SJ1001T_2      SJ000026
## DYE1680:AAACCCAGTAAGAACT SJRHB000026_C5-SJ1001T_2      SJ000026
##                               Preferred_Sample_ID.1 Sample_Type_Code.1
## DYE1680:AAACCCAAGAGTCAGC      SJ1001T_2      C
## DYE1680:AAACCCAAGATACCAA      SJ1001T_2      C
## DYE1680:AAACCCAAGCTTCTAG      SJ1001T_2      C
## DYE1680:AAACCCACACTGGATT      SJ1001T_2      C
## DYE1680:AAACCCACAGAAGTGC      SJ1001T_2      C
## DYE1680:AAACCCAGTAAGAACT      SJ1001T_2      C
##                               Prior_Comp_Bio_ID.1 PATH.1 NOTES.1 match_SJID_RPMI.1
## DYE1680:AAACCCAAGAGTCAGC      SJRHB000026      <NA>      NA      SJRHB000026_no
## DYE1680:AAACCCAAGATACCAA      SJRHB000026      <NA>      NA      SJRHB000026_no
## DYE1680:AAACCCAAGCTTCTAG      SJRHB000026      <NA>      NA      SJRHB000026_no
## DYE1680:AAACCCACACTGGATT      SJRHB000026      <NA>      NA      SJRHB000026_no
## DYE1680:AAACCCACAGAAGTGC      SJRHB000026      <NA>      NA      SJRHB000026_no
## DYE1680:AAACCCAGTAAGAACT      SJRHB000026      <NA>      NA      SJRHB000026_no
##                               ID_with_sample_swap.1 SAMPLE_with_sample_swap.1
## DYE1680:AAACCCAAGAGTCAGC      DYE1680      2664137_DYE1680
## DYE1680:AAACCCAAGATACCAA      DYE1680      2664137_DYE1680
## DYE1680:AAACCCAAGCTTCTAG      DYE1680      2664137_DYE1680
## DYE1680:AAACCCACACTGGATT      DYE1680      2664137_DYE1680
## DYE1680:AAACCCACAGAAGTGC      DYE1680      2664137_DYE1680
## DYE1680:AAACCCAGTAAGAACT      DYE1680      2664137_DYE1680
##
## DYE1680:AAACCCAAGAGTCAGC /research/groups/dyergrp/projects/dyergrp_hartwell/common/illum

```

```

1/2-2261003/
## DYE1680:AAACCCAAGATACCAA /research/groups/dyergrp/projects/dyergrp_hartwell/common/illum
1/2-2261003/
## DYE1680:AAACCCAAGCTTCTAG /research/groups/dyergrp/projects/dyergrp_hartwell/common/illum
1/2-2261003/
## DYE1680:AAACCCACACTGGATT /research/groups/dyergrp/projects/dyergrp_hartwell/common/illum
1/2-2261003/
## DYE1680:AAACCCACAGAAGTGC /research/groups/dyergrp/projects/dyergrp_hartwell/common/illum
1/2-2261003/
## DYE1680:AAACCCAGTAAGAACT /research/groups/dyergrp/projects/dyergrp_hartwell/common/illum
1/2-2261003/
##
## S.Score.1 G2M.Score.1 Phase.1 UMAP_1.x UMAP_2.x
## DYE1680:AAACCCAAGAGTCAGC -0.1902376 -0.3227688 G1 -1.4770168 9.253440
## DYE1680:AAACCCAAGATACCAA -0.1182429 -0.2969047 G1 -3.4309131 9.242017
## DYE1680:AAACCCAAGCTTCTAG -0.2180694 -0.3405832 G1 -1.3636737 5.705616
## DYE1680:AAACCCACACTGGATT -0.2376908 -0.3945082 G1 -2.9635712 9.708883
## DYE1680:AAACCCACAGAAGTGC -0.1505212 -0.1332644 G1 -1.0462393 7.168882
## DYE1680:AAACCCAGTAAGAACT 0.4046701 -0.1800174 S 0.4893034 7.784715
##
## PC_1.x PC_2.x PC_3.x PC_4.x PC_5.x
## DYE1680:AAACCCAAGAGTCAGC -16.39490 -0.06040939 8.992026 -5.282849 2.5461725
## DYE1680:AAACCCAAGATACCAA -16.64311 -0.38841290 8.531777 -4.574383 -0.1577565
## DYE1680:AAACCCAAGCTTCTAG -17.02470 -0.26009097 9.657486 -4.846276 2.6670944
## DYE1680:AAACCCACACTGGATT -15.89784 0.65302425 9.408316 -4.259913 -0.3211939
## DYE1680:AAACCCACAGAAGTGC -15.76642 0.20835809 7.609702 -3.459250 2.9684272
## DYE1680:AAACCCAGTAAGAACT -11.64230 1.66521080 7.193732 -2.240468 2.9785559
##
## PC_6.x PC_7.x PC_8.x PC_9.x
## DYE1680:AAACCCAAGAGTCAGC 6.56743106 5.4881562 4.7613897 -1.7865470
## DYE1680:AAACCCAAGATACCAA 3.63397381 -0.4869838 -0.6133165 0.4781322
## DYE1680:AAACCCAAGCTTCTAG 8.97802837 4.5605539 2.0849652 -1.6105814
## DYE1680:AAACCCACACTGGATT -0.05474395 -2.2903917 3.7300108 -0.5776052
## DYE1680:AAACCCACAGAAGTGC 9.52270257 4.2045560 0.8241562 -1.4506361
## DYE1680:AAACCCAGTAAGAACT 7.74266906 4.9746429 -0.1754950 0.8894317
##
## PC_10.x PC_11.x PC_12.x PC_13.x PC_14.x
## DYE1680:AAACCCAAGAGTCAGC 0.4089495 0.5652035 -0.3806822 -1.011628 1.4937597
## DYE1680:AAACCCAAGATACCAA 1.9618030 1.2614855 1.0912301 -1.861864 2.1052791
## DYE1680:AAACCCAAGCTTCTAG 2.0010723 0.1003519 0.4319290 -2.877628 3.0628591
## DYE1680:AAACCCACACTGGATT 1.5921441 -1.2365924 -0.7285866 -2.319632 2.5677598
## DYE1680:AAACCCACAGAAGTGC 0.7336166 0.9973878 0.6110972 -1.937815 0.2880151
## DYE1680:AAACCCAGTAAGAACT -1.4764380 -1.9228858 -3.1487889 2.355300 0.5375213
##
## PC_15.x PC_16.x PC_17.x PC_18.x
## DYE1680:AAACCCAAGAGTCAGC -0.6740777 -1.26045048 -1.1002345 5.6047590
## DYE1680:AAACCCAAGATACCAA 2.1245733 0.53280848 -2.1309847 6.0946147
## DYE1680:AAACCCAAGCTTCTAG 0.6736039 0.05249832 -1.3541124 4.7995531
## DYE1680:AAACCCACACTGGATT -1.4860827 -1.62879851 -2.3296588 4.5801502
## DYE1680:AAACCCACAGAAGTGC -0.3293910 1.34307035 -0.6013391 -0.8125743
## DYE1680:AAACCCAGTAAGAACT 0.9685769 -2.24292698 0.8368750 -3.2915569
##
## PC_19.x PC_20.x PC_21.x PC_22.x PC_23.x
## DYE1680:AAACCCAAGAGTCAGC 2.8967472 2.553915 -1.1964531 -1.7446081 -
0.38584092
## DYE1680:AAACCCAAGATACCAA -0.1098369 1.302813 0.5558706 0.5823816 0.36094618
## DYE1680:AAACCCAAGCTTCTAG 1.0072936 1.547666 -0.3104649 -0.8433231 0.18960913

```



```

## DYE1680:AAACCCACACTGGATT 1.3088686 1.924711 -1.7640707 -2.5096479 0.08471221
## DYE1680:AAACCCACAGAAGTGC -0.5978964 -2.268368 0.8100166 0.8282191 0.05591339
## DYE1680:AAACCCAGTAAGAACT -0.9768113 0.543290 0.8136991 2.9906515 -
0.51044236
##
PC_24.x PC_25.x PC_26.x PC_27.x PC_28.x
## DYE1680:AAACCCAAGAGTCAGC 0.26184242 0.2345830 2.535177 0.55438094 5.646553
## DYE1680:AAACCCAAGATACCAA 0.07283511 -2.0171602 -2.480327 -0.14655514 -
5.756900
## DYE1680:AAACCCAAGCTTCTAG 0.51470577 -1.2350765 1.248190 1.13687684 2.915157
## DYE1680:AAACCCACACTGGATT 0.20947787 -1.5374189 2.780055 0.00924215 2.388767
## DYE1680:AAACCCACAGAAGTGC 0.60514027 -0.2262589 -1.149257 2.78306784 -
2.389567
## DYE1680:AAACCCAGTAAGAACT -0.59741445 5.0010707 -0.746392 -1.64070414 -
5.421385
##
PC_29.x PC_30.x RNA_snn_res.0.10
## DYE1680:AAACCCAAGAGTCAGC 3.754707 -1.2801447 1
## DYE1680:AAACCCAAGATACCAA -4.257351 1.1886556 1
## DYE1680:AAACCCAAGCTTCTAG 1.994136 -0.1955019 1
## DYE1680:AAACCCACACTGGATT 1.642289 0.0246024 1
## DYE1680:AAACCCACAGAAGTGC -3.795585 3.8918452 1
## DYE1680:AAACCCAGTAAGAACT -6.199011 1.9952317 1
##
seurat_clusters.1 RNA_snn_res.0.1.1 RNA_snn_res.0.2.1
## DYE1680:AAACCCAAGAGTCAGC 2 1 1
## DYE1680:AAACCCAAGATACCAA 58 1 1
## DYE1680:AAACCCAAGCTTCTAG 2 1 1
## DYE1680:AAACCCACACTGGATT 2 1 1
## DYE1680:AAACCCACAGAAGTGC 29 1 1
## DYE1680:AAACCCAGTAAGAACT 124 1 1
##
RNA_snn_res.0.3.1 RNA_snn_res.0.4.1 RNA_snn_res.0.5.1
## DYE1680:AAACCCAAGAGTCAGC 8 8 9
## DYE1680:AAACCCAAGATACCAA 8 8 9
## DYE1680:AAACCCAAGCTTCTAG 1 1 9
## DYE1680:AAACCCACACTGGATT 8 8 9
## DYE1680:AAACCCACAGAAGTGC 1 8 9
## DYE1680:AAACCCAGTAAGAACT 1 10 8
##
RNA_snn_res.0.6.1 RNA_snn_res.0.7.1 RNA_snn_res.0.8.1
## DYE1680:AAACCCAAGAGTCAGC 7 8 8
## DYE1680:AAACCCAAGATACCAA 7 8 8
## DYE1680:AAACCCAAGCTTCTAG 7 8 8
## DYE1680:AAACCCACACTGGATT 7 8 8
## DYE1680:AAACCCACAGAAGTGC 7 8 8
## DYE1680:AAACCCAGTAAGAACT 10 11 13
##
RNA_snn_res.0.9.1 RNA_snn_res.1.1 RNA_snn_res.2.1
## DYE1680:AAACCCAAGAGTCAGC 9 10 27
## DYE1680:AAACCCAAGATACCAA 9 10 13
## DYE1680:AAACCCAAGCTTCTAG 9 10 27
## DYE1680:AAACCCACACTGGATT 9 10 27
## DYE1680:AAACCCACAGAAGTGC 9 10 13
## DYE1680:AAACCCAGTAAGAACT 13 12 11
##
RNA_snn_res.3.1 RNA_snn_res.4.1 RNA_snn_res.5.1
## DYE1680:AAACCCAAGAGTCAGC 26 28 17

```

## DYE1680:AAACCCAAGATACCAA	20	16	9		
## DYE1680:AAACCCAAGCTTCTAG	26	28	17		
## DYE1680:AAACCCACACTGGATT	26	28	17		
## DYE1680:AAACCCACAGAAGTGC	39	44	52		
## DYE1680:AAACCCAGTAAGAACT	39	44	62		
##	RNA_snn_res.6.1	RNA_snn_res.7.1	RNA_snn_res.8.1		
## DYE1680:AAACCCAAGAGTCAGC	15	11	4		
## DYE1680:AAACCCAAGATACCAA	7	3	80		
## DYE1680:AAACCCAAGCTTCTAG	15	11	4		
## DYE1680:AAACCCACACTGGATT	15	11	4		
## DYE1680:AAACCCACAGAAGTGC	52	47	51		
## DYE1680:AAACCCAGTAAGAACT	82	95	104		
##	RNA_snn_res.9.1	RNA_snn_res.10.1	UMAP_1.y	UMAP_2.y	
## DYE1680:AAACCCAAGAGTCAGC	6	2	5.2027514	2.8551131	
## DYE1680:AAACCCAAGATACCAA	57	58	4.8270543	-	
1.0893938					
## DYE1680:AAACCCAAGCTTCTAG	6	2	4.7985137	1.9250976	
## DYE1680:AAACCCACACTGGATT	6	2	4.9758880	3.0762058	
## DYE1680:AAACCCACAGAAGTGC	32	29	6.1563537	-	
0.1735911					
## DYE1680:AAACCCAGTAAGAACT	95	124	0.9281711	-	
0.7314588					
##	PC_1.y	PC_2.y	PC_3.y	PC_4.y	PC_5.y
## DYE1680:AAACCCAAGAGTCAGC	-14.99294	-9.743418	-7.6353402	2.9332377	-
0.78601409					
## DYE1680:AAACCCAAGATACCAA	-16.71809	-5.952063	-0.3889715	-0.4360290	0.74538672
## DYE1680:AAACCCAAGCTTCTAG	-16.03057	-11.262267	-6.1506574	-0.1809578	1.48210689
## DYE1680:AAACCCACACTGGATT	-17.07094	-3.207242	1.5985109	5.3802679	-
0.08968461					
## DYE1680:AAACCCACAGAAGTGC	-14.15228	-13.241235	-7.0030156	-1.6916667	1.26167535
## DYE1680:AAACCCAGTAAGAACT	-11.74029	-10.606897	-1.1069191	-4.8273344	-
0.85038172					
##	PC_6.y	PC_7.y	PC_8.y	PC_9.y	PC_10.y
## DYE1680:AAACCCAAGAGTCAGC	0.8604809	-1.52804117	-2.269825	-6.2100177	5.5451535
## DYE1680:AAACCCAAGATACCAA	-4.1223143	0.03496484	-3.886041	-8.4302986	1.6027635
## DYE1680:AAACCCAAGCTTCTAG	-0.1122444	-2.73946011	-4.512047	-6.1647699	2.4779725
## DYE1680:AAACCCACACTGGATT	-1.0709175	-3.69640104	-2.446371	-6.2184213	3.4936762
## DYE1680:AAACCCACAGAAGTGC	0.5816906	-2.16792623	-1.246656	1.3938973	-
1.7077229					
## DYE1680:AAACCCAGTAAGAACT	9.3624741	1.56704327	1.889982	0.9006061	-
0.2264416					
##	PC_11.y	PC_12.y	PC_13.y	PC_14.y	
## DYE1680:AAACCCAAGAGTCAGC	2.0235296	-0.5767616	3.01791901	1.30408019	
## DYE1680:AAACCCAAGATACCAA	-0.2725346	1.1886442	0.07801078	-2.04972977	
## DYE1680:AAACCCAAGCTTCTAG	1.1227242	-0.7895150	2.40985189	0.03991787	
## DYE1680:AAACCCACACTGGATT	4.5696957	1.2777738	3.10694325	0.75590098	
## DYE1680:AAACCCACAGAAGTGC	-0.3244840	-0.5918644	0.76139781	-5.86771913	
## DYE1680:AAACCCAGTAAGAACT	-2.9652965	0.6425002	-8.84879571	-8.16437660	
##	PC_15.y	PC_16.y	PC_17.y	PC_18.y	
## DYE1680:AAACCCAAGAGTCAGC	-4.0261631	7.0106319	3.6541063	-2.11460651	
## DYE1680:AAACCCAAGATACCAA	3.9976860	-6.2201414	-5.8508662	3.73510922	

```

## DYE1680:AAACCCAAGCTTCTAG 0.5638844 2.7247101 3.0695968 -0.08425802
## DYE1680:AAACCCACACTGGATT -3.4819508 5.1005356 -0.8068963 3.14229130
## DYE1680:AAACCCACAGAAGTGC 2.8490743 -4.2627948 -0.0953835 4.10963963
## DYE1680:AAACCCAGTAAGAACT -1.0179600 0.2157179 -6.5586051 -5.22702710
##
## PC_19.y PC_20.y PC_21.y PC_22.y PC_23.y
## DYE1680:AAACCCAAGAGTCAGC 2.0278977 -2.9440294 3.754152 1.9816658 3.7540959
## DYE1680:AAACCCAAGATACCAA 0.2770261 1.1128306 -2.913314 -1.1717716 2.0221170
## DYE1680:AAACCCAAGCTTCTAG -0.5117608 -0.9633634 -2.590465 2.2979075 1.7908833
## DYE1680:AAACCCACACTGGATT 1.1195503 -1.9697480 4.291697 -0.8560660 4.5850100
## DYE1680:AAACCCACAGAAGTGC -3.9793333 -1.6880893 -2.487131 0.5550594 -
0.6072706
## DYE1680:AAACCCAGTAAGAACT -1.4071959 -4.2818247 5.603335 -0.8642200 -
4.6408268
##
## PC_24.y PC_25.y PC_26.y PC_27.y
## DYE1680:AAACCCAAGAGTCAGC 0.1053464 -4.3349781 0.4235789 -0.62341869
## DYE1680:AAACCCAAGATACCAA -2.6261339 2.1426604 -1.8239533 2.32107169
## DYE1680:AAACCCAAGCTTCTAG -1.0343722 -0.1068329 -0.4314079 -0.03960706
## DYE1680:AAACCCACACTGGATT -2.8609561 -2.7089397 1.4632763 -2.62936463
## DYE1680:AAACCCACAGAAGTGC 0.1967018 0.5439947 -0.1861994 0.27194799
## DYE1680:AAACCCAGTAAGAACT -0.9441400 -0.5417910 -2.6711955 -3.09020315
##
## PC_28.y PC_29.y PC_30.y
## DYE1680:AAACCCAAGAGTCAGC -0.8653674 -0.1693271 -0.1355869
## DYE1680:AAACCCAAGATACCAA 1.9695339 -0.3511901 1.2764081
## DYE1680:AAACCCAAGCTTCTAG 0.8889436 -0.3044911 0.6352865
## DYE1680:AAACCCACACTGGATT 3.4284315 1.7298663 1.3980331
## DYE1680:AAACCCACAGAAGTGC -0.1152079 -0.2278947 -5.5162817
## DYE1680:AAACCCAGTAAGAACT 0.8391258 0.1272049 2.4948336

```

```

saveRDS(seurat_obj, file = paste0(results_dir, "/", glue::glue("seurat_obj_integrated_{inte

```

10 Session Info

```
## R version 4.4.0 (2024-04-24)
## Platform: x86_64-pc-linux-gnu
## Running under: Ubuntu 22.04.4 LTS
##
## Matrix products: default
## BLAS: /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
## LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblas-p-r0.3.20.so; LAPACK vers
##
## locale:
## [1] C
##
## time zone: Etc/UTC
## tzcode source: system (glibc)
##
## attached base packages:
## [1] grid      stats      graphics  grDevices  utils      datasets  methods
## [8] base
##
## other attached packages:
## [1] knitr_1.49          reshape2_1.4.4      SeuratWrappers_0.2.0
## [4] RcppPlanc_1.0.0     rliger_2.1.0.9004   harmony_1.2.3
## [7] Rcpp_1.0.14         future_1.34.0       RColorBrewer_1.1-3
## [10] ggthemes_5.1.0      patchwork_1.3.0     scooter_0.0.0.9004
## [13] SeuratObject_5.0.2  Seurat_4.4.0        lubridate_1.9.4
## [16] forcats_1.0.0       stringr_1.5.1       dplyr_1.1.4
## [19] purrr_1.0.2         readr_2.1.5         tidyr_1.3.1
## [22] tibble_3.2.1        ggplot2_3.5.1       tidyverse_2.0.0
## [25] yaml_2.3.10
##
## loaded via a namespace (and not attached):
## [1] jsonlite_1.8.9      magrittr_2.0.3      ggbeeswarm_0.7.2
## [4] spatstat.utils_3.1-2 farver_2.1.2         rmarkdown_2.29
## [7] ragg_1.3.3          vctrs_0.6.5         ROCR_1.0-11
## [10] spatstat.explore_3.3-4 tinytex_0.54         htmltools_0.5.8.1
## [13] sass_0.4.9          sctransform_0.4.1   parallelly_1.42.0
## [16] bslib_0.9.0         KernSmooth_2.23-26  htmlwidgets_1.6.4
## [19] ica_1.0-3           plyr_1.8.9          cachem_1.1.0
## [22] plotly_4.10.4       zoo_1.8-12          igraph_2.1.4
## [25] mime_0.12           lifecycle_1.0.4     pkgconfig_2.0.3
## [28] rsvd_1.0.5          Matrix_1.7-2        R6_2.5.1
## [31] fastmap_1.2.0       fitdistrplus_1.2-2  shiny_1.10.0
## [34] digest_0.6.37       colorspace_2.1-1    S4Vectors_0.44.0
## [37] tensor_1.5          irlba_2.3.5.1       textshaping_1.0.0
## [40] labeling_0.4.3      progressr_0.15.1    spatstat.sparse_3.1-0
## [43] timechange_0.3.0    httr_1.4.7          polyclip_1.10-7
## [46] abind_1.4-8         compiler_4.4.0      remotes_2.5.0
## [49] bit64_4.6.0-1       withr_3.0.2         MASS_7.3-64
## [52] ggsci_3.2.0         tools_4.4.0         vipor_0.4.7
## [55] lmtest_0.9-40       beeswarm_0.4.0      httpuv_1.6.15
```

## [58] future.apply_1.11.3	goftest_1.2-3	glue_1.8.0
## [61] nlme_3.1-167	promises_1.3.2	Rtsne_0.17
## [64] cluster_2.1.8	generics_0.1.3	gtable_0.3.6
## [67] spatstat.data_3.1-4	tzdb_0.4.0	data.table_1.16.4
## [70] hms_1.1.3	sp_2.2-0	BiocGenerics_0.52.0
## [73] spatstat.geom_3.3-5	RcppAnnoy_0.0.22	ggrepel_0.9.6
## [76] RANN_2.6.2	pillar_1.10.1	spam_2.11-1
## [79] vroom_1.6.5	later_1.4.1	splines_4.4.0
## [82] lattice_0.22-6	survival_3.8-3	bit_4.5.0.1
## [85] deldir_2.0-4	tidyselect_1.2.1	miniUI_0.1.1.1
## [88] pbapply_1.7-2	gridExtra_2.3	scattermore_1.2
## [91] RhpcBLASctl_0.23-42	stats4_4.4.0	xfun_0.50
## [94] matrixStats_1.5.0	stringi_1.8.4	lazyeval_0.2.2
## [97] evaluate_1.0.3	codetools_0.2-20	BiocManager_1.30.25
## [100] cli_3.6.3	uwot_0.2.2	xtable_1.8-4
## [103] reticulate_1.40.0	systemfonts_1.2.1	jquerylib_0.1.4
## [106] munsell_0.5.1	globals_0.16.3	spatstat.random_3.3-2
## [109] png_0.1-8	ggtrastr_1.0.2	spatstat.univar_3.1-1
## [112] parallel_4.4.0	dotCall64_1.2	listenv_0.9.1
## [115] viridisLite_0.4.2	scales_1.3.0	ggiridges_0.5.6
## [118] leiden_0.4.3.1	crayon_1.5.3	rlang_1.1.5
## [121] cowplot_1.1.3		